

Temporal-Support Identifiability in Event-Detection Evaluation Pipelines

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Abstract

001 *Event-detection benchmarks compare a prediction with a*
002 *label at one nominal timestamp, which stands in for sev-*
003 *eral non-equivalent temporal quantities: when the evidence*
004 *supporting the prediction was acquired, which instant its*
005 *output represents, and over what interval the label was ac-*
006 *quired. Released artifacts partially specify temporal se-*
007 *mantics, but a nominal benchmark timestamp does not by*
008 *itself encode the effective predictor timing, the recurrent*
009 *context represented by a pooled score, or the attribution of*
010 *within-exposure event timing.*

011 *A single reported event-detection score can pool predic-*
012 *tions made with materially different temporal support. In*
013 *the released SSM-ViT evaluation pipeline, recurrent his-*
014 *tory varies from one to twenty-one windows within the*
015 *same pooled score. A paired intervention that changes only*
016 *the chunk boundary evaluates the same frames with short*
017 *versus long recurrent history and changes S5-B by 4.20*
018 *mAP over all labeled boxes and 5.23 mAP on the velocity-*
019 *evaluable subset, while keeping the frames, weights, la-*
020 *bels, and evaluator fixed. We call the resulting ambiguity*
021 *temporal-support identifiability: whether the released pro-*
022 *cedure and artifacts uniquely determine the temporal quanti-*
023 *ties represented by a reported score. On the predictor side,*
024 *the newest RVT input window has an ablation-sensitivity*
025 *centroid 23.81 ms before the label instant, whereas the*
026 *regression-based temporal coefficient is -2.40 ± 7.18 ms*
027 *under the reported specification and is not interpreted as a*
028 *direct estimate of the network's recurrent-support centroid.*
029 *On the label side, within-exposure event timing contains an*
030 *illumination-locked component, so attributing that timing*
031 *uniquely to object motion requires separating the illumina-*
032 *tion component. Together, these audits separate recover-*
033 *able artifact facts from temporal quantities left implicit by*
034 *the reported scores.*

1. Introduction

An object detection benchmark evaluates a prediction $\hat{s}$ against a ground truth s^* through a function $\rho(\hat{s}, s^*)$ of position, shape and class, indexed by a query time t . The query time selects which label the prediction is compared against. Beyond selecting the label at query time t , the scoring function contains no variable representing temporal support or effective state time, so a temporal discrepancy is scored as displacement. Neither side's temporal support is specified by that nominal instant. A predictor computes $\hat{s}$ from a window of sensor data. A label is produced by an exposure, an annotator or tracker working on the exposed image, and a geometric transfer into another sensor's frame.

Some of these quantities are constrained by released artifacts rather than by the metric itself. The detector configuration fixes an input window, the preprocessing script in the detector's own repository places that window's end on the label it is scored against, and the published exposure files fix an acquisition interval, while the scoring function reads none of them. The released artifacts therefore determine some temporal quantities and leave others under-specified. The distinction can be tested directly.

We distinguish quantities recoverable from released artifacts from quantities explicitly represented by the reported benchmark score. A temporal descriptor may be reconstructable from released artifacts while remaining unreported or heterogeneous within a pooled benchmark score. The released artifacts partially specify temporal support: input and exposure intervals are reconstructable, whereas effective predictor timing, recurrent context represented by a pooled score, and within-exposure attribution are not uniquely encoded by the nominal timestamp. We formulate this ambiguity as benchmark temporal-support identifiability: whether the released protocol and artifacts uniquely determine the temporal quantities represented by a reported score. Three temporal quantities are audited, each from released artifacts alone.

Predictor input and carried support. The event branch of the detector examined here [21] consumes a window

of 50 ms in ten bins whose end sits on the label instant (Sec. 3.2), so under uniform weighting its evidence is centered 25 ms earlier. The trained network’s bin sensitivities are not uniform, and ablating each bin in turn on real validation samples puts the ablation-sensitivity centroid at -23.81 ms (Sec. 3.3). Over a 1000 ms horizon, the newest window accounts for 22.7% of the summed window-ablation sensitivity, while the remaining 77.3% lies in preceding windows. Its regression-based temporal coefficient, estimated on nominal-time-matched detections under Eq. 2 with a vector regression, a placebo term and a sign-flip control on the velocity estimator, is -2.40 ± 7.18 ms, statistically compatible with zero under the selected specification, and it sits 26.2 ms from that centroid under the reported specification. Sensitivity to the input and the coefficient of Eq. 2 are distinct operational quantities. The nominal timestamp does not by itself encode either quantity (Secs. 3.2–3.5).

Evaluation support. For the released SSM-ViT implementation, the nominally carried state is inert across evaluation chunks, making the available recurrent history a deterministic function of chunk position. Chunk position assigns detections between one and twenty-one windows of recurrent history under a single reported score, and the score moves with it: on all labeled boxes, the 1–4 versus 17–21 window contrast is 5.32 mAP points net of the RVT control, while the same-frame chunk-boundary intervention changes S5-B by 4.20 points (5.23 points on the velocity-evaluable subset). See Sec. 3.3.

Label acquisition support. Across the eighteen DSEC training sequences whose exposure files resolve, the published exposure width varies by a factor of 127 against a frame period fixed at 50 ms, with 32.5 % of frames pinned at an auto-exposure ceiling in six sequences (Fig. 3). Those sequences carry the harmonic series of a 100 Hz mains flicker, and between 95.4 % and 100.0 % of labeled boxes carry more power at the fundamental than at an off-frequency arm on the same events. Thus, within-exposure event timing contains illumination-locked structure and therefore cannot be uniquely attributed to object timing without separating that structure (Secs. 4–4.4).

Identifiability under the metric is then read in the metric’s own units. On the velocity-evaluable subset, displacing ground-truth centers across the measured label-to-centroid interval changes mAP by 0.04 points and does not reorder the five released checkpoints. Separately, over all labeled boxes, the nominal validation-split ordering is reproduced in only 42.5 % of sequence-bootstrap replicates. These results place the observed displacement effect at the benchmark’s sampling-resolution floor. Table 2 relates this resolution to the benchmark’s speed composition (Supplement Secs. 11–12).

Accepted work on event-detection timing takes the

benchmark’s temporal index as given and improves what is measured against it: sensor latency is calibrated away [22, 26], compute latency is folded into average precision [6, 27, 32], and the temporal window is redesigned [21, 23]. We formalize the remaining question as benchmark temporal-support identifiability: whether the released protocol and artifacts uniquely determine the temporal quantities represented by a reported score. The clearest failure occurs in streaming evaluation. In the released SSM-ViT pipeline, one pooled score combines detections with one to twenty-one windows of recurrent history, and changing only the chunk boundary shifts S5-B by 4.20 mAP on the same labeled frames. The predictor- and label-side audits then show two additional distinctions hidden by the nominal timestamp: input sensitivity is not equivalent to an output-time coefficient, and within-exposure event timing cannot be attributed uniquely to object motion without separating illumination-locked structure.

2. Related work

Sensor and runtime latency. Prior work calibrates sensor latency [22, 26], incorporates runtime latency into average precision [6, 27, 32], or proposes temporal tolerance [33]. These methods address the time from an event to an output. Our question is whether a released benchmark specification uniquely determines the temporal quantities behind its score.

Temporal windows and benchmark identifiability. RVT and SSM-ViT redesign or parameterize temporal windows [21, 23], while adaptive-window methods compare alternative partitions [25]. Event-RGB frequency mismatch and exposure duration have also been noted [19, 20, 34]. The closest accepted work above calibrates sensor or runtime latency or redesigns temporal windows, but does not audit whether a released benchmark protocol uniquely specifies predictor support, output-time interpretation, and label-acquisition support.

Benchmark reliability and evaluation analysis. Prior work studies detector calibration, stratified model evaluation, label-induced benchmark instability, and event-vision evaluation protocols [28–31]. These analyses examine uncertainty or error in reported performance, whereas our focus is the temporal support and acquisition semantics that a nominal detection timestamp leaves implicit.

Label time and attribution. Waymo and annotation studies estimate per-object capture or annotation offsets [2–4]. Event datasets expose per-box time fields, but their degrees of freedom are not necessarily specified. LET-3D-AP

and TIDE/DETAD/HOTA decompose or bucket spatial error while taking the label clock as given [1, 8–11].

Event benchmarks and exposure. Gen1, 1 Mpx, DSEC-Det and Ev-3DOD use asynchronous measurements, rectified frames, or interpolated labels [13, 16–18]. LEOD establishes causal input windows [24]. The present work reads the released window and exposure artifacts as protocol evidence and tests whether the resulting temporal quantities are uniquely specified.

3. Predictor-side temporal descriptors: input sensitivity and output-time coefficient

3.1. Identifiability and the timing estimands

A benchmark releases one tuple per evaluated instant: a nominal timestamp, the labels asserted at it, the predictions scored against them, and the score. Three temporal quantities are latent in that tuple rather than recorded in it: the temporal support of the predictor’s input and carried state, the effective state time its output refers to, and the acquisition support of the labels. Call such a quantity *protocol-identifiable* when the released protocol and artifacts determine it uniquely. We distinguish quantities recoverable from released artifacts from quantities explicitly represented by the reported benchmark score. A temporal descriptor may be reconstructable from released artifacts while remaining unreported or heterogeneous within a pooled benchmark score. The released artifacts partially specify temporal support: input and exposure intervals are reconstructable, whereas effective predictor timing, recurrent context represented by a pooled score, and within-exposure attribution are not uniquely encoded by the nominal timestamp. The measurements below quantify distinct temporal descriptors that are collapsed onto the same nominal benchmark timestamp.

Let a predictor P emit a state at query time t . A signed effective-weight model uses bin centers c_k and weights a_k with $\sum_k a_k = 1$, giving the effective timestamp offset $\tau_P = \sum_k a_k c_k$ under constant velocity. The empirical ablation profile instead assigns non-negative sensitivities $s_k \geq 0$ to bins and has centroid $c_s = \sum_k s_k c_k / \sum_k s_k$. These are distinct quantities: zero-ablation sensitivity depends on both the predictor and the input, and is not in general a decomposition of temporal weights. Sections 3.2–3.3 measure this operational input-sensitivity profile over the newest window. Section 3.4 estimates the regression coefficient τ under Eq. 2. Interpreting $-\tau$ as an effective state-time offset requires the absence of residual velocity-proportional spatial bias. Section 3.5 compares $-c_s$ with the regression coefficient τ , equivalently c_s with the regression-implied timestamp $\tau_P = -\tau$.

3.2. The window and the instant it is indexed by

Two released files fix the input side. The dataset configuration `config/dataset/gen1.yaml` of the detector’s own repository [21] names the representation `stacked_histogram_dt=50_nbins=10`, a window of 50 ms divided into ten bins of 5 ms. Line 405 of `scripts/genx/preprocess_dataset.py` constructs the array `ev_repr_timestamps_us_end` by counting backwards from a label timestamp in steps of the window duration, and extends it between consecutive label timestamps. Each representation is therefore indexed by the end of its window, and those ends are placed on label times. A window attached to a label at time t holds the events of $[t - 50 \text{ ms}, t]$, whose uniform weighting centroid falls 25 ms before the instant t at which the label asserts a state. Both statements are read from the released source, and the 25 ms figure is a reference point rather than a property of the trained network.

3.3. The measured influence profile

The newest-window ablation-sensitivity centroid is computed from the measured bin-ablation sensitivities of a released checkpoint. We take 480 samples of the released Gen1 validation split with the released `rvt-t` checkpoint, warm the recurrent state for eight steps, and ablate each bin in turn. For sample i , let Y_i denote the emitted detection tensor from the unablated input and $Y_i^{(-k)}$ the tensor obtained after zeroing temporal bin k while keeping the incoming recurrent state identical. We define the ablation sensitivity, or influence, of bin k as

$$s_k = \frac{1}{N} \sum_{i=1}^N \frac{\|Y_i^{(-k)} - Y_i\|_2}{\|Y_i\|_2}, \quad N = 480, \quad (1)$$

$$c_s = \frac{\sum_k s_k c_k}{\sum_k s_k},$$

where c_k is the center time of bin k relative to the label instant. The ablated forward pass receives the same incoming recurrent state as the corresponding reference pass, and its returned state is discarded. The profile is a property of the current window and not of a perturbed history.

The profile is balanced between the two halves of the window (Fig. 1a) and yields an ablation-sensitivity centroid of -23.81 ms , compared with -25.00 ms for uniform weighting. The released configuration therefore places the measured centroid near the window reference without implying uniform network sensitivity. Across zero, mean, cross-sample replacement, and gradient instruments, the centroid spans -26.35 to -22.35 ms (Supplement Sec. 2). The reported centroid applies to the deterministic 12-sequence subset described in the supplement. This is the current window’s window-ablation sensitivity profile

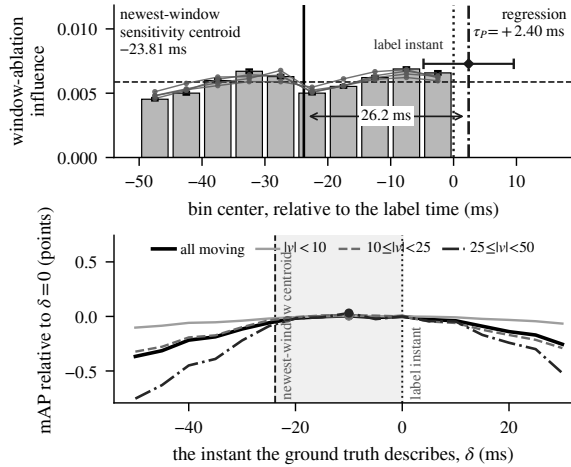


Figure 1. The two quantities shown are non-equivalent operational summaries, not competing estimates of a common latent timestamp. Their point estimates are 26.2 ms apart. Panel (b) shows that average precision is nearly invariant over the measured 23.81 ms label-to-centroid displacement. The 26.2 ms separation is not an estimate of architectural latency. (a) Window-ablation influence of each of the ten bins of the released $rvt-t$ input window, over 480 Gen1 validation samples with the recurrent state warmed eight steps, with sample-level standard errors. Error bars are descriptive standard errors across the sampled instances and are not used for inference. Uncertainty of the ablation-sensitivity centroid is evaluated by sequence bootstrap in Supplement Sec. 2. Dashed: the mean over bins. Dotted: the label instant. Solid: the ablation-sensitivity centroid. Dash-dot, with its standard error shaded: the regression-implied timestamp under Eq. 2, $\tau_p = -\tau$, of Sec. 3.4. Gray: the four other checkpoints, each rescaled to its own mean. (b) Average precision against the temporal center displacement δ applied to the ground-truth boxes, by object speed in px/s, each curve relative to its own value at the label instant, with its maximum marked. The shaded band runs from the centroid to the label instant, spanning 0.61 px at the ninetieth percentile of label speed against an annotation dispersion of 0.6353 px (Supplement Sec. 3). The mAP span is 0.37 points over all moving boxes and 0.78 points in the 25–50 px/s stratum, where the centroid displacement costs 0.08 points.

with the recurrent state held fixed, not a decomposition of the full temporal support.

Recurrent support beyond the newest window. Over a 1000 ms horizon, the newest window accounts for 22.7% of the summed window-ablation sensitivity, while the remaining 77.3% lies in preceding windows. The horizon centroid shifts from -161 ms at 500 ms to -299 ms at 1000 ms, so the measured range does not determine a converged full-history centroid. Supplement Sec. 14 characterizes the tail and its fill controls.

Across the released checkpoints and a second architecture. RVT releases three Gen1 checkpoints differing only in embedding width, attention head dimension and FPN depth. SSM-ViT [23] releases two more in which each stage’s ConvLSTM is replaced by an S5 state space layer, at

9.68 and 18.19 M parameters against RVT’s 9.87 and 18.54. It distributes RVT’s own preprocessed Gen1 and inherits its window construction. On the same sequences, and driving each the way its own release drives it, all 5 checkpoints put the newest window’s ablation-sensitivity centroid between -23.76 and -24.72 ms, none further than 1.24 ms, or 25 % of one bin, from the uniform-weight value.

Recurrent history under the released streaming evaluation.

The state the SSM release carries between evaluation chunks reaches none of its outputs: it enters as a multiplicative factor on the first coefficient of an associative scan whose outputs never read that coefficient. Under the released streaming evaluation this bounds the available recurrent history to the current 21-window chunk, so the history behind a detection ranges from one window to 21 with its position in the chunk and the reported number averages over all 21. Reconstructing that position from the release’s own index files over 20 296 labeled frames and scoring every labeled box, S5-B reaches 34.445 mAP on the frames given one to four windows and 40.226 on those given seventeen to 21 (Fig. 2). Differencing against the three RVT checkpoints, evaluated on the same frames with a state that crosses chunk boundaries, controls for frame-position effects shared with the RVT checkpoints and leaves 5.32 ± 0.53 points for S5-B ($z = 10.0$) and 4.37 ± 0.61 for S5-S ($z = 7.2$), against placebo differences among the RVT checkpoints of at most 0.67 points ($z = 1.56$). Permuting chunk position within each sequence puts both SSM values more than 7.6 standard deviations outside their own null and every placebo inside it. A label-free statistic, mean detection confidence, separates the same two groups in the same direction (Supplement Sec. 13). A paired within-frame chunk-boundary intervention keeps the target frame, weights, labels, and evaluator fixed. Shifting every chunk boundary 5 windows later carries the starved block onto positions 16–19, so the 3 574 affected frames are scored twice, under one to four windows of history and under seventeen to twenty, with the same weights and labels. Paired within frame, the long-history arm gains 5.23 ± 0.49 mAP points for S5-B and 4.81 ± 0.52 for S5-S, with 0.0579 ± 0.0048 and 0.0509 ± 0.0035 of mean detection confidence. For S5-B, it yields 5.23 ± 0.49 mAP points on the velocity-evaluable subset and 4.20 ± 0.36 points over every labeled box. Fig. 2 draws one such pair. Restricting evaluation to frames receiving 17–21 windows yields an S5-B score 1.75 points above the pooled released-evaluation score. The support behind that window is not fixed in the same way either: over 576 samples of the three RVT capacities, the newest window’s share of a 1000 ms horizon falls from 22.7 % to 19.0 % and the sensitivity centroid moves from -299 to -339 ms.

3.4. Regression-based temporal coefficient

The influence profile is a statement about the network’s input. Even a predictor with uniform local sensitivity can emit a state describing a later instant, and that quantity is measured from the same checkpoint’s own detections.

A detector reporting the state an object occupied τ earlier incurs the physical displacement $-\tau v$, where v is the latent true velocity. We observe only the finite-differenced label velocity $\tilde{v} = v + \eta$ and therefore fit Eq. 2 using $\tilde{v}$. Thus $\tau = -\tau_P$ on the axis of Sec. 3. With $\hat{c}$ the predicted center, c^* the label center, A the area of the ground-truth box, $\hat{u} = \tilde{v}/\|\tilde{v}\|$ and R a quarter turn, b is a two-dimensional intercept, fitted globally or as per-sequence intercepts in the reported specification,

$$\hat{c} - c^* = b - \tau \tilde{v} + \rho R\tilde{v} + \gamma\sqrt{A}\hat{u} + \varepsilon \quad (2)$$

is fitted by stacking both components of every match into one least squares problem, with standard errors clustered by sequence. It is the temporal calibration of [5] carried to a box center, taken as a slope rather than as the per-object ratio $-\langle \hat{c} - c^*, \hat{u} \rangle / \|\tilde{v}\|$, whose variance diverges as its denominator approaches zero. For comparison, the per-object ratio evaluated on the same rows has a mean of $+24.22 \pm 27.73$ ms, on the hypothesis evaluated in Sec. 3.5, with an interquartile range of 430 ms. Written this way, Eq. 2 fits the two-dimensional residual directly with $\tilde{v}$ as a regressor. Unlike the per-object ratio, it does not first project the residual onto $\hat{u}$ and divide by $\|\tilde{v}\|$. The reported full specification adds four coordinate/size regressors and per-sequence intercepts to the terms displayed in Eq. 2. The coefficient is a temporal offset only if no velocity-proportional spatial bias remains after these controls. Such a bias is indistinguishable from τ in Eq. 2.

$R\tilde{v}$ is the label velocity rotated a quarter turn. Because a pure temporal offset acts along $\tilde{v}$ rather than $R\tilde{v}$, ρ serves as a negative-control coefficient. A nonzero value indicates residual spatial misspecification. $\sqrt{A}\hat{u}$ absorbs a displacement along travel proportional to box side, which is needed because that residual grows with box size and box size correlates with label speed at $+0.139$ on this split.

The finite difference used for the velocity. $\tilde{v}$ is finite differenced from labels. Label noise enters forward and backward velocity differences with opposite shared-noise signs. The centered difference excludes the current label center and removes this direct same-frame term. Supplement Secs. 1 and 4 derive and calibrate this effect, so the reported regression uses the centered estimator.

Regression dataset and matching. Rows are detections matched to nominal-time ground truth at $\text{IoU} \geq 0.3$ with centered label velocities available, leaving 22 534 matches

Specification	τ (ms)	placebo ρ (ms)
<i>velocity estimator, no geometric controls</i>		
forward difference	-14.68 ± 4.99	-6.53 ± 1.68
backward difference	$+45.65 \pm 5.59$	-6.15 ± 1.71
centered difference	$+17.51 \pm 5.06$	-7.76 ± 1.92
<i>controls added, centered difference throughout</i>		
+ box side along travel	$+14.62 \pm 5.47$	-7.75 ± 1.91
+ four geometric terms	-7.62 ± 7.01	-4.12 ± 1.90
+ per-sequence bias	-2.40 ± 7.18	-2.66 ± 1.71

Table 1. The regression-based temporal coefficient τ of the released `rvt-t` checkpoint over 22 534 matched boxes of 376 Gen1 validation sequences, and the placebo ρ beside it, the coefficient on the rotated label velocity, which no temporal offset can reach. Standard errors are clustered by sequence. The forward and backward control fits are reported in Supplement Sec. 1.

from 376 sequences. The estimate is conditional on this matched subset. Supplement Sec. 5 reports association-rule sweeps, including overlap-free matching.

Table 1 is a specification ladder, and the specification was chosen by the placebo rather than by τ . Under Eq. 2 with only the box-side term, $\tau = +14.62 \pm 5.47$ ms, but ρ is -7.75 ms at 4.0 standard errors, which indicates a term omitted from Eq. 2 that $R\tilde{v}$ is a proxy for. A split by the sign of the horizontal velocity locates it: ρ changes sign across that split, at -27.99 ms and $+21.84$ ms, which a rotational term would not do. Most label motion here is horizontal, so $R\tilde{v}$ maps a horizontal speed onto a vertical residual. A bias differing between oncoming and same-direction traffic is represented by this rotated regressor. Entering box height and image row along $\hat{y}$ and box width and image column along $\hat{x}$, and then a constant vector per sequence, returns the placebo to -2.66 ms at 1.6 standard errors. That is the specification reported, and the criterion that selected it is a coefficient on a regressor orthogonal to $\tilde{v}$.

Under the reported specification, $\tau = -2.40 \pm 7.18$ ms, with a sequence-bootstrap 95% interval of $[-17.0, +11.5]$ ms. Across geometric-control specifications, τ spans -20.2 to $+11.1$ ms. Because the reported specification was selected using the orthogonal placebo coefficient, its nominal standard error does not include specification-selection uncertainty. We therefore treat the point estimate descriptively and use the full specification sweep to assess robustness.

3.5. Relating input-sensitivity timing to the regression-based temporal coefficient

The comparison combines measurements under different recurrent-state conditions. The centroid uses an eight-step warm-up with the incoming state fixed during ablation. The regression uses detections from the validation stream with the released state carried across frames. These are operational timing measurements of the same checkpoint, not estimates of a common full-recurrent-support centroid. The

26.2 ms numerical separation is descriptive only: the two quantities are distinct operational summaries and are not competing estimates of a common latent timestamp. Under the reported specification and its annotation-error assumptions, the point-estimate difference is 26.2 ms. The regression coefficient has a standard error of 7.2 ms. This is a regression-coefficient result rather than a latency estimate of this architecture. Expressed on Eq. 2's sign convention, the newest-window centroid corresponds numerically to $\tau = +23.81$ ms. This comparison relates two operational estimands and does not assume that Eq. 2 must recover the ablation centroid. Thus, 23.81 ms is the measured interval between the label instant and the newest-window ablation-sensitivity centroid, whereas 26.2 ms is the point-estimate difference between that centroid and the reported regression coefficient. The measured horizons do not determine a converged full-history centroid. The standard error quoted is the output-time term's. The variation across specifications is substantial relative to the reported standard error and is not captured by that standard error.

The detector is trained against the ground truth at the label time on a window ending there, so a measured offset near zero is consistent with the training objective indexed at the label time. The objective does not enforce the interval, and the instant the output describes and the centroid of the evidence used to produce it are different quantities that one benchmark timestamp stands for. At the ninetieth percentile of label speed on this split, 25.7 px/s, the 23.81 ms is 0.61 px of displacement, comparable to the annotation dispersion of the labels themselves (0.6353 px), and at the ninety-ninth it is 1.29 px. That ablation-sensitivity centroid and the regression-based temporal coefficient are measurable from released artifacts alone.

Effect on benchmark mAP. This experiment is not a test of whether temporal support matters in general. It asks whether Gen1's particular speed composition makes the measured RVT interval visible to its current mAP. The benchmark metric can be evaluated in its own units: we translate each ground-truth box center by $\delta\tilde{v}$, keeping its size and class fixed. This provides a first-order surrogate for temporal displacement under local constant velocity. The released predictions remain untouched. A detection carries no velocity of its own. Over 20 296 frames and 40 698 boxes, 68.7% of which carry a centered velocity and the rest are ignored. Ground-truth boxes without a centered velocity are marked ignored during evaluation, and detections matched to ignored boxes are excluded from TP/FP accounting. The resulting mAP against δ is smooth with its maximum at -10 ms (Fig. 1b). Along the axis of Eq. 2 that is $\tau = +10$ ms, and mAP varies by at most 0.04 points over the interval. This descriptive curve does not by itself identify a preferred temporal value.

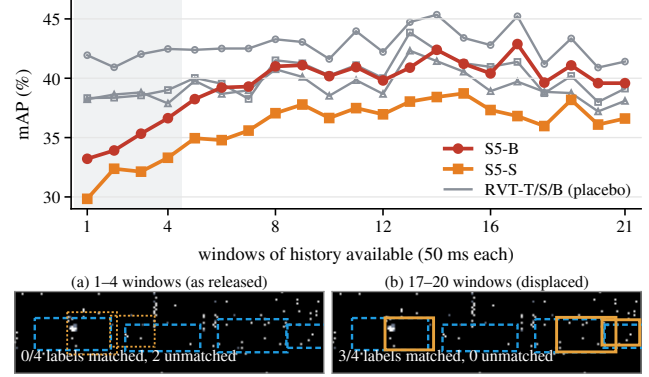


Figure 2. **Top:** benchmark score against the recurrent history the released streaming evaluation made available, at each of the 21 chunk positions, over every labeled Gen1 box, no displacement applied. Shaded: the one-to-four-window block the release scores together with the rest. Between that block and the frames given seventeen windows or more, the two SSM checkpoints rise by 5.78 and 4.83 points and the three RVT checkpoints, scored on the same frames with state that crosses chunk boundaries, by at most 0.80. **Bottom:** one frame of the same-frame intervention, S5-B. Both panels share the frame, the labels (dashed) and the weights, and only the chunk boundary moves. Solid: a detection matched to a label of its class at $\text{IoU} \geq 0.5$. Dotted: unmatched. Drawn above confidence 0.3 and scored at every confidence. The example is illustrative only. All paired quantitative estimates use all 3 574 frames. Supplement Sec. 13.

The full -60 to $+60$ ms sweep moves mAP by 0.9 points. On the -50 to $+30$ ms grid summarized in Table 2, the all-moving span is 0.37 points. Displacing the ground truth to the window's centroid costs 0.0004 absolute mAP (0.04 percentage points, 0.12% relative to 0.3346). Average precision on the velocity-evaluable subset is nearly invariant over the measured label-to-centroid displacement, and at the common -23.81 ms displacement the ordering of all 5 released checkpoints is unchanged on the validation split. Scoring each checkpoint at its own preferred displacement likewise leaves the validation-split ordering unchanged in the reported strata. Both statements are bounded by the stability of the ordering under sequence resampling at this scale. The smallest timing difference that would reorder an adjacent pair is 43.7 ms over all moving boxes, $1.83\times$ the measured interval, and 11.2 ms in the 25–50 px/s stratum. Separately, on every labeled box rather than the velocity-evaluable subset the displacement acts on, the benchmark's own cluster resampling holds the validation-split ordering in only 42.5% of replicates, its tightest adjacent pair separated by 0.050 points against a paired standard error of 0.53. A common displacement sweep reveals one localized inversion between $\text{rvt}-b$ and S5-S in the 25–50 px/s stratum. Table 2 locates the flatness in that composition: 72.1% of the moving boxes travel below 10 px/s, where the displacement is 0.05 px, and the sweep deepens to 0.78 points between 25 and 50 px/s, where it is 0.77 px, past the labels' own annotation dispersion of 0.6353 px (Supplement

speed (px/s)	share (%)	med. speed	centroid (px)	Δ mAP	span
all moving	100.0	4.0	0.10	0.04	0.37
< 10	72.1	2.1	0.05	0.02	0.11
10–25	19.2	15.0	0.36	0.01	0.33
25–50	7.5	32.3	0.77	0.08	0.78
≥ 50	1.2	57.4	1.37	0.00	0.37

Table 2. The sweep of Fig. 1b by object speed over the 27 943 ground-truth boxes with centered velocity. Other boxes are ignored. Δ mAP is the cost of shifting the ground truth to the newest-window ablation-sensitivity centroid at -23.81 ms. Span is the mAP range over the -50 to $+30$ ms sweep. Both values are reported in points.

Sec. 3). The near invariance therefore characterizes Gen1’s metric resolution, not the absence of a temporal discrepancy.

4. A complementary within-exposure attribution limit

4.1. Exposure and labels

The predictor-side analysis concerns the temporal support of released detector inputs and outputs. DSEC provides a separate exposure-side analysis: published exposure intervals bound the interval within which event timing is observed, while within-exposure event timing contains illumination-locked structure and therefore cannot be uniquely attributed to object timing without separating that structure. DSEC [12] publishes per-frame exposure metadata, and the files resolve for eighteen train sequences. The width varies by a factor of 127 against a frame period fixed at 50 ms, and within one recording by up to a factor of 12.5. The auto-exposure ceiling of $14\,996\,\mu\text{s}$ holds 32.5 % of frames, in six of the eighteen sequences, and the distribution is not continuous over that range (Fig. 3). This motivates treating ceiling exposure as a distinct stratum.

The released DSEC-Det labels comprise 390 118 boxes on 6693 tracks across 60 sequences. Every label sits on the frame clock, and the median spacing between consecutive label times is $50\,001\,\mu\text{s}$. Over 361 628 triples of consecutive labels on a track, 7.1 % of middle box centers fall within a hundredth of a pixel of their neighbors’ linear interpolant. This rules out exact local linear interpolation as a general description of the released trajectories without identifying the underlying annotation process. DSEC-Det ships more than one annotation version [15], the supplement names the one measured here. Median box-center image speed is 40 px/s, with 360 px/s at the ninety-ninth percentile.

4.2. Mains-frequency signature and the per-pixel phase test

Swiss mains is 50 Hz, and rectified lighting can produce the tested 100 Hz modulation. The pre-specified 100-Hz line and its harmonics appear in the event-rate spectrum of every

ceiling sequence, while the daytime control lacks the same pattern (Fig. 4). Per-pixel phase tests and the pre-specified frequency control support the same illumination signature.

Per pixel, the Rayleigh statistic $Z = R^2/N$ is $\text{Exp}(1)$ under independent uniform phases. The off-frequency and daytime controls empirically calibrate this nominal null, while the ceiling sequence has median $Z = 2.935$ at 100 Hz and 24.07 % of live pixels reach the nominal $p < 10^{-3}$ threshold under the independent-uniform-phase reference. The illumination attribution therefore relies primarily on the frequency-matched and daytime controls rather than on the nominal Rayleigh reference alone.

4.3. All six ceiling sequences, per labeled box

The same statistic taken per labeled box, over the events inside that box during that frame’s own exposure window, extends to every ceiling sequence. Across the six sequences and 16 516 boxes, 95.4–100.0% satisfy the pre-specified criterion in Supplement Table 2.

4.4. Within-box temporal dispersion under that signature

That signature limits attribution of within-exposure event timing to object timing. The per-box 137-Hz arm is used as a same-event descriptive control rather than as an independent spectral-resolution test. The illumination attribution is supported primarily by the long-slice 100-Hz harmonic structure and the daytime control. Within-box dispersion is $206.6\,\mu\text{s}$ against a null of $93.0\,\mu\text{s}$, with an excess of $183.6\,\mu\text{s}$. A frame-level permutation preserving pooled temporal distributions and box counts gives a null of $78.6\,\mu\text{s}$, and the excess remains after the clean-pixel control. Random-position boxes show that the dispersion is not specific to annotated object regions (Supplement Sec. 9), reinforcing an attribution-limit interpretation rather than an object-motion interpretation.

5. Scope and limitations

Each result is conditional on a stated instrument: local window-ablation sensitivity, nominal-time association under the reported regression specification, and a local constant-velocity displacement surrogate. The SSM chunk-boundary result applies to the two released SSM-ViT checkpoints under their released streaming evaluator. Gen1 and DSEC are separate case studies, and no cross-dataset universality is claimed. All Gen1 quantitative claims are validation-split claims obtained from ported released checkpoints, and sequence bootstrap claims use the 406 sequences represented in the stored-detection evaluation. Accordingly, our quantitative conclusions concern validation-split comparisons and the behavior of the ported released checkpoints.

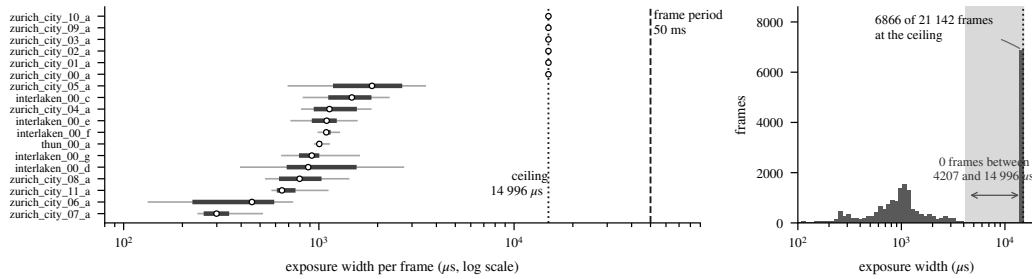


Figure 3. Published per-frame exposure width across the eighteen DSEC training sequences whose exposure files resolve: per sequence (left, log axis, median, interquartile range and 5–95 % band, against the fixed 50 ms frame period) and pooled over 21 142 frames (right). 32.5 % of frames sit at the auto-exposure ceiling of 14 996 μs and zero fall between the largest non-ceiling width of 4207 μs and that ceiling, separating the two exposure regimes. The six ceiling sequences are those measured in Sec. 4.

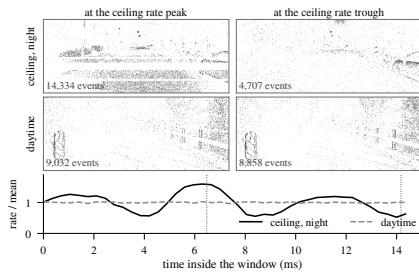


Figure 4. Top: 350 μs slices of the ceiling-exposure sequence `zurich.city_09_a` and of the daytime control, at the peak and trough of the ceiling sequence's own event rate inside its densest published exposure window. Bottom: that rate in 350 μs bins, each sequence normalized to its own mean. The ceiling rate moves by a factor of 3.0 between its extremes, the daytime rate by 1.1.

6. Conclusion

Released artifacts only partially specify a nominal timestamp's temporal semantics. In the released SSM-ViT pipeline, predictions with one to twenty-one recurrent windows are pooled into one score, and a same-frame intervention that changes only chunk position shifts S5-B by 4.20 mAP while keeping the frames, weights, labels, and evaluator fixed. RVT's input sensitivity and regression-based output-time coefficient are distinct descriptors, while DSEC's within-exposure timing contains illumination-locked structure requiring separation before attribution to object motion. These results motivate reporting predictor support and label acquisition intervals.

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